LONDON DATA WEEK 2026 · HACKATHON

Can You Vouch For It?

Evaluating an Artificial Intelligence (AI) enrichment pipeline for 20,685 London datasets — *honestly.*

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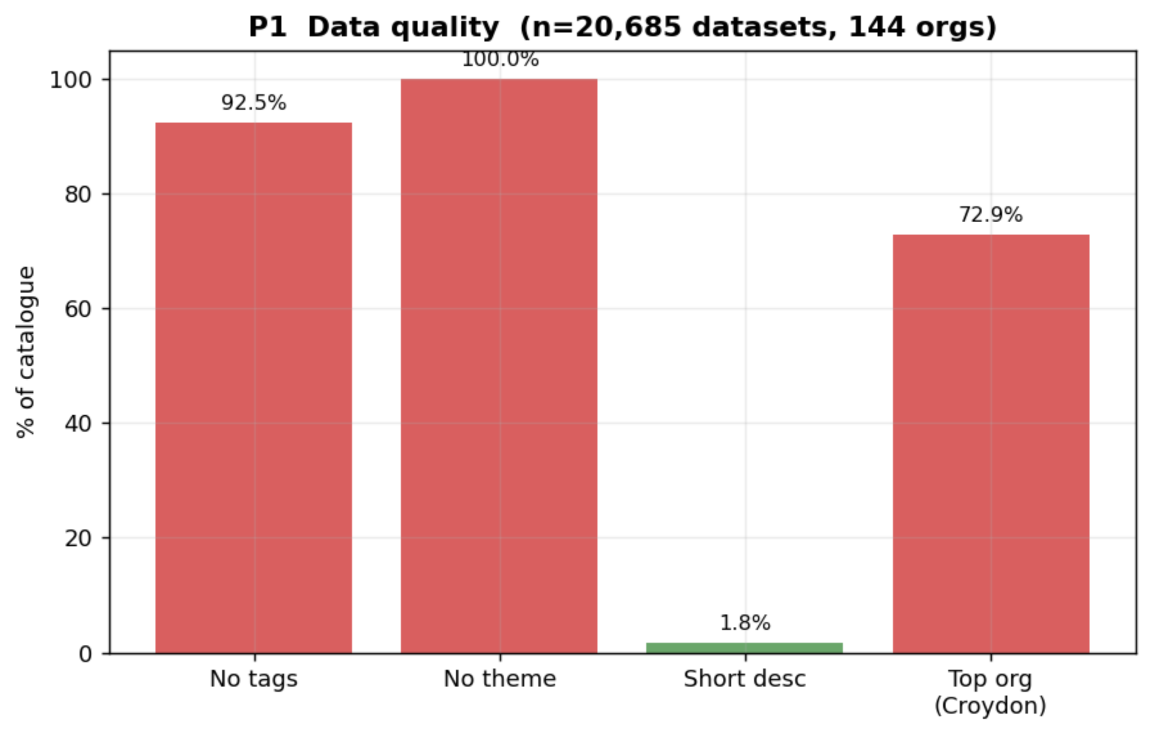
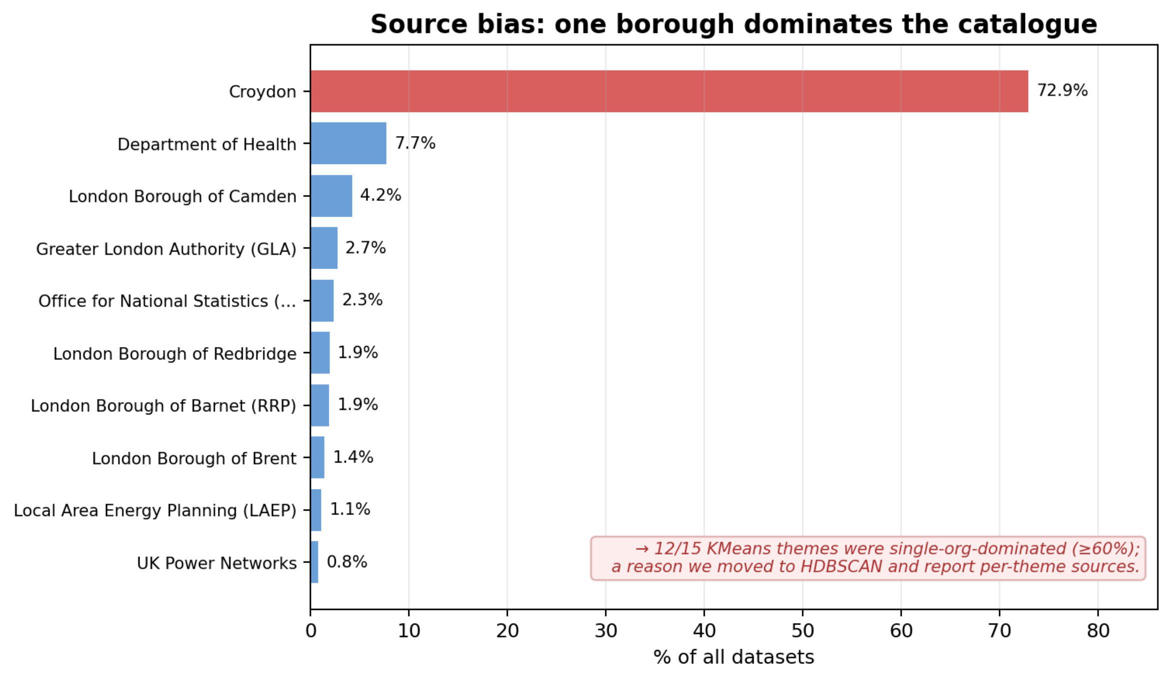
Problem 3 · Discovering London's Data

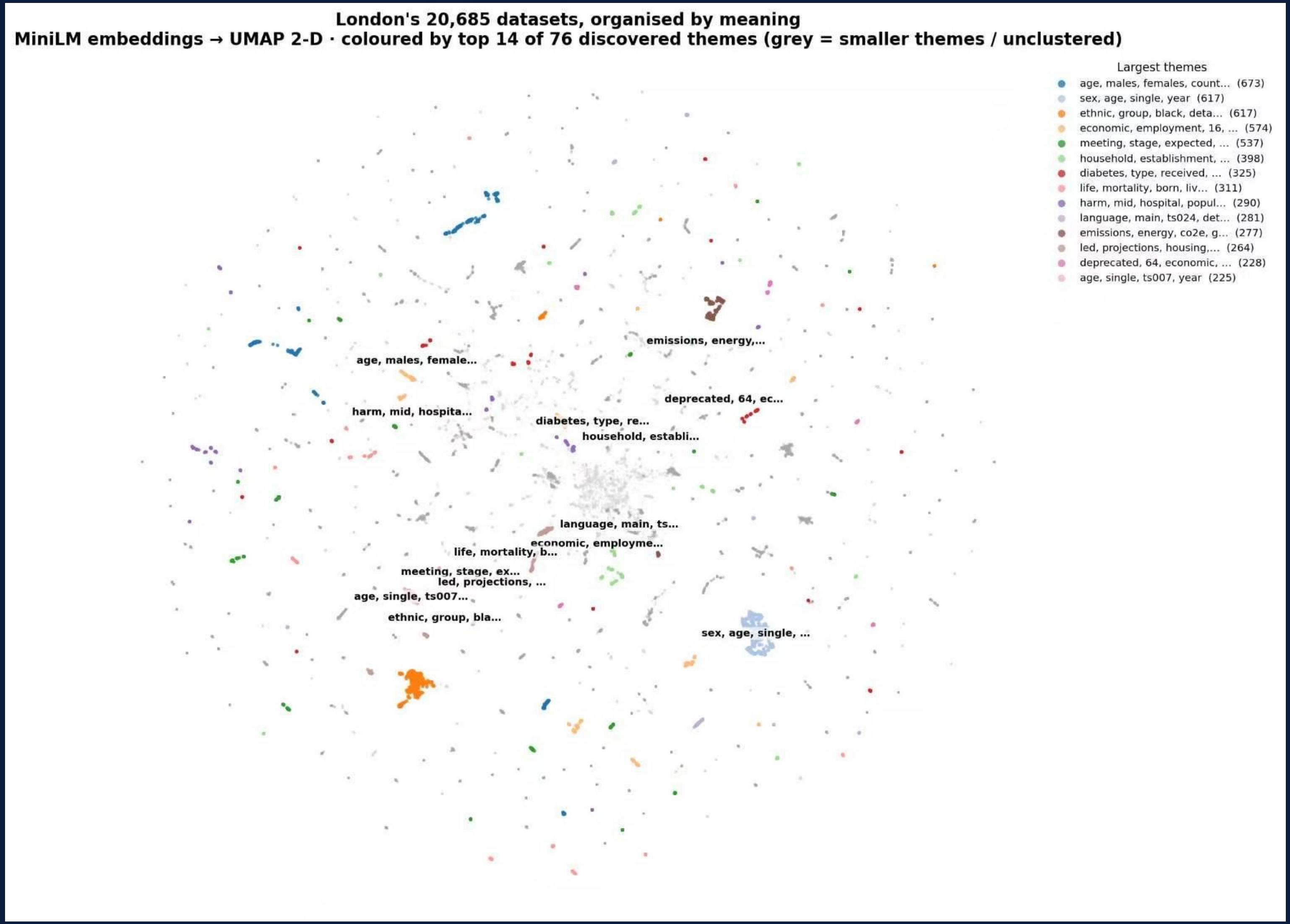
Good Data, Buried Deep





20,685 datasets. 92.5% have no tags, 100% have no theme, and one borough (Croydon) supplies 72.9% of everything — so today's search is really just a search of Croydon.

Turning Text Into Meaning



76

themes discovered automatically — no k chosen by hand

How it works

Each dataset's title + notes become a 384-dimension MiniLM vector. Similar meaning = nearby vectors. HDBSCAN groups the dense areas into themes — grey dots are honestly left unclustered.

Search quality: two baselines vs. our method

Same query · same corpus (title + notes) · Top-5 results · 20,685 London datasets

Query: "young people not in work"

Synonym / acronym recall (NEET)

#	Keyword Today's Library · word-count	TF-IDF Fair lexical baseline	Semantic Ours · MiniLM embeddings
1	London Labour Market, Skills and Employment Indicators ~	Young People ~	Percentage of male 16 and 17 year olds not in education, employment or training... ✓
2	TS058 - Distance travelled to work Works mainly from home rate ✗	Children & Young People ~	Number of male 16 and 17 year olds not in education, employment or training... ✓
3	TS058 - Distance travelled to work Works mainly from home ✗	Young people who have attained a full level 3 qualification by age 19 ~	Percentage of female 16 and 17 year olds not in education, employment or training... ✓
4	TS058 - Distance travelled to work Works mainly at an offshore installation, in... ✗	New analysis on autism in Camden: children and young people ~	Number of female 16 and 17 year olds not in education, employment or training... ✓
5	33kV Circuit Operational Data Half Hourly - South Eastern Power Networks... ✗	Children And Young People With Special Education Needs And Disabilities ~	16 to 17 year olds not in education, employment or training (NEET) or whose... ✓

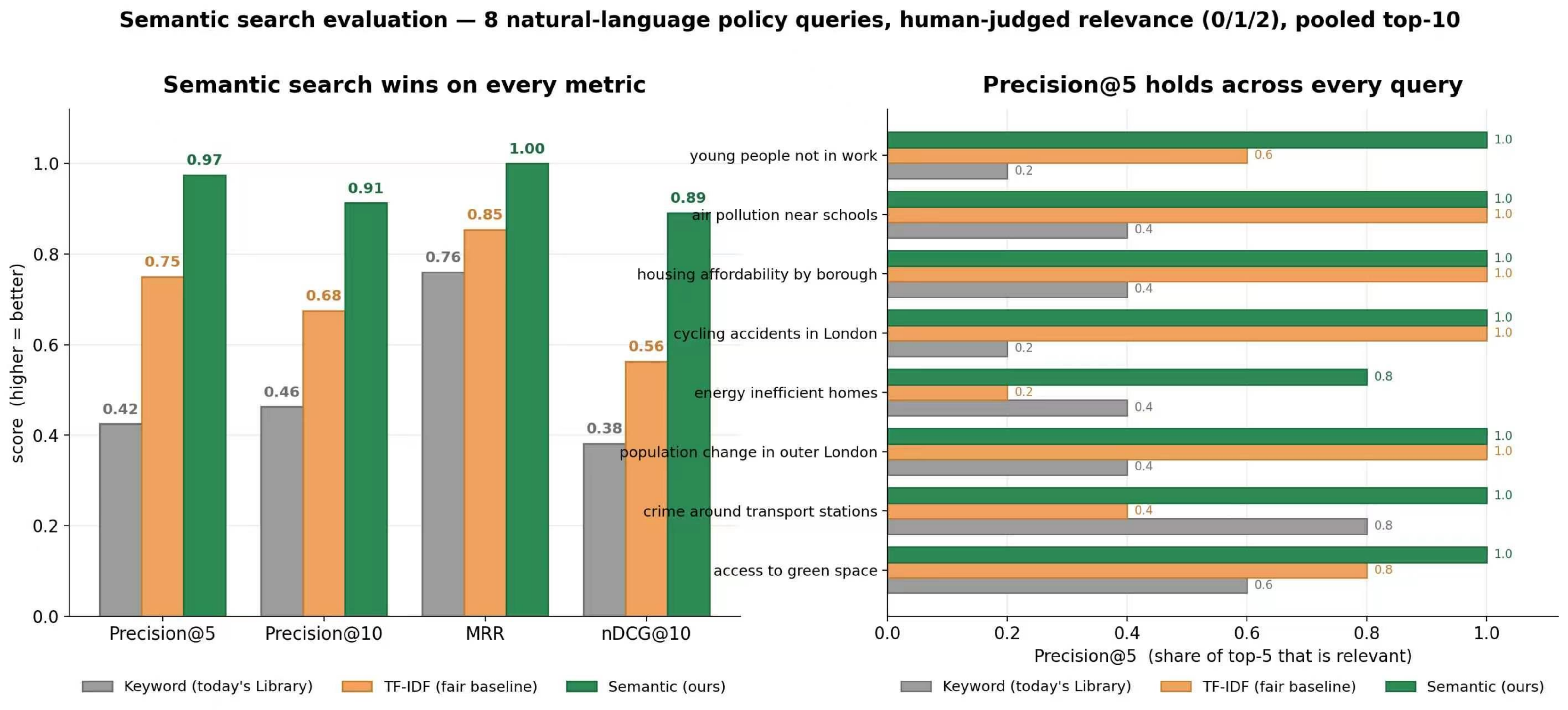
✓ Relevant ~ Partial match ✗ Off-topic

Query: "air quality and pollution"

Robustness to long-text noise (COVID)

#	Keyword Today's Library · word-count	TF-IDF Fair lexical baseline	Semantic Ours · MiniLM embeddings
1	Coronavirus (COVID-19) Vaccine Roll Out ✗	Camden Air Quality Monitoring ✓	Analysing Air Pollution Exposure in London ✓
2	Coronavirus (COVID-19) Deaths ✗	Analysing Air Pollution Exposure in London ✓	Air pollution: estimated fraction of mortality attributable to particulate... ✓
3	London Atmospheric Emissions Inventory (LAEI) 2019 ✓	London Air Quality Network Camden ✓	Air pollution: fine particulate matter (new method - concentrations of total... ✓
4	Coronavirus (COVID-19) Weekly Update ✗	London Air Quality Network Camden Latest Week ✓	Understanding Health Impacts of Air Pollution in London ✓
5	Public Health Outcomes Framework Indicators ~	Low Emission Neighbourhoods ✓	Fraction of mortality attributable to particulate air pollution - Persons -... ✓

Keyword mimics today's Library · TF-IDF is a fair lexical baseline · embeddings recover synonyms (NEET) and resist long-text noise (COVID)



8 real policy questions · human-judged relevance · four industry-standard IR metrics

Are the Auto-Generated Tags Any Good?

We manually reviewed a random sample of 30 datasets and their auto-tags.

93.3%

of sampled tags were specific and usable as suggestions

15.8%

of all tags contain stray digits (e.g. '32', '500') — noise to filter

7.7%

of tags are purely numeric — should be dropped automatically

1.43

average words per tag — tags run a little too short/fragmented

dataset_idx	title	notes_snippet	tag	heuristic	human
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o	retrofit	Good	Good
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o	workflow	Good	Good
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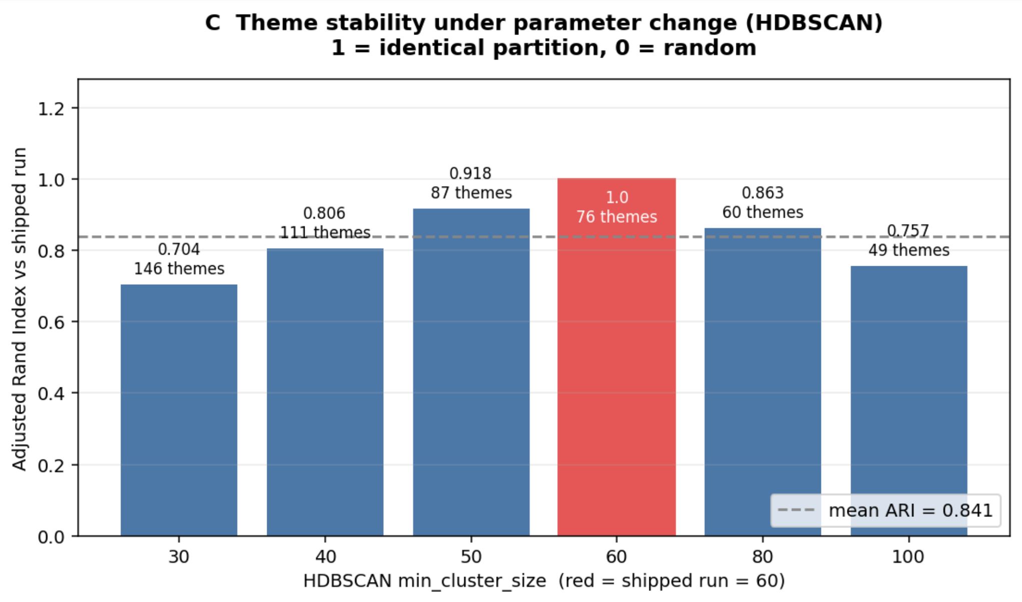
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o	diagram	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	number eligible	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	eligible pupils	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	additional	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	english	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	eligible	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	language	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	language eu	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	european language	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	services	Okay	Okay
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	european	Good	Good
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	housing led	Okay	Okay
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	led	Okay	Okay
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	75 84	Bad	Bad
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	population projections	Good	Good
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. Th	projections	Okay	Okay
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	18	Bad	Bad
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	18 years	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	aged 18	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	years persons	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. Th	persons 18	Good	Good
5270	AF006: stroke risk assessed w. CHA2DS2-VASc (den.incl.exc.) - Persons - All ages (Denominator	Note from Data for London: Croydon Observatory provided access to this dataset. Th	vasc	Good	Good
5270	AF006: stroke risk assessed w. CHA2DS2-VASc (den.incl.exc.) - Persons - All ages (Denominator	Note from Data for London: Croydon Observatory provided access to this dataset. Th	risk	Good	Good
5270	AF006: stroke risk assessed w. CHA2DS2-VASc (den.incl.exc.) - Persons - All ages (Denominator	Note from Data for London: Croydon Observatory provided access to this dataset. Th	stroke	Good	Good
5270	AF006: stroke risk assessed w. CHA2DS2-VASc (den.incl.exc.) - Persons - All ages (Denominator	Note from Data for London: Croydon Observatory provided access to this dataset. Th	assessed	Good	Good
5270	AF006: stroke risk assessed w. CHA2DS2-VASc (den.incl.exc.) - Persons - All ages (Denominator	Note from Data for London: Croydon Observatory provided access to this dataset. Th	ages denominator	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. Th	59	Bad	Bad
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. Th	population	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. Th	population estimates	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. Th	aged 16	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. Th	population aged	Good	Good
5891	DEPRECATED: Smoking status at time of delivery - Female - All ages	Note from Data for London: Croydon Observatory provided access to this dataset. Th	smoking status	Good	Good
5891	DEPRECATED: Smoking status at time of delivery - Female - All ages	Note from Data for London: Croydon Observatory provided access to this dataset. Th	delivery	Good	Good

Verdict: tags are good enough to suggest, not good enough to auto-publish. A simple digit-filter removes most of the noise before it reaches a human reviewer.

Do the Themes Make Sense to a Human?

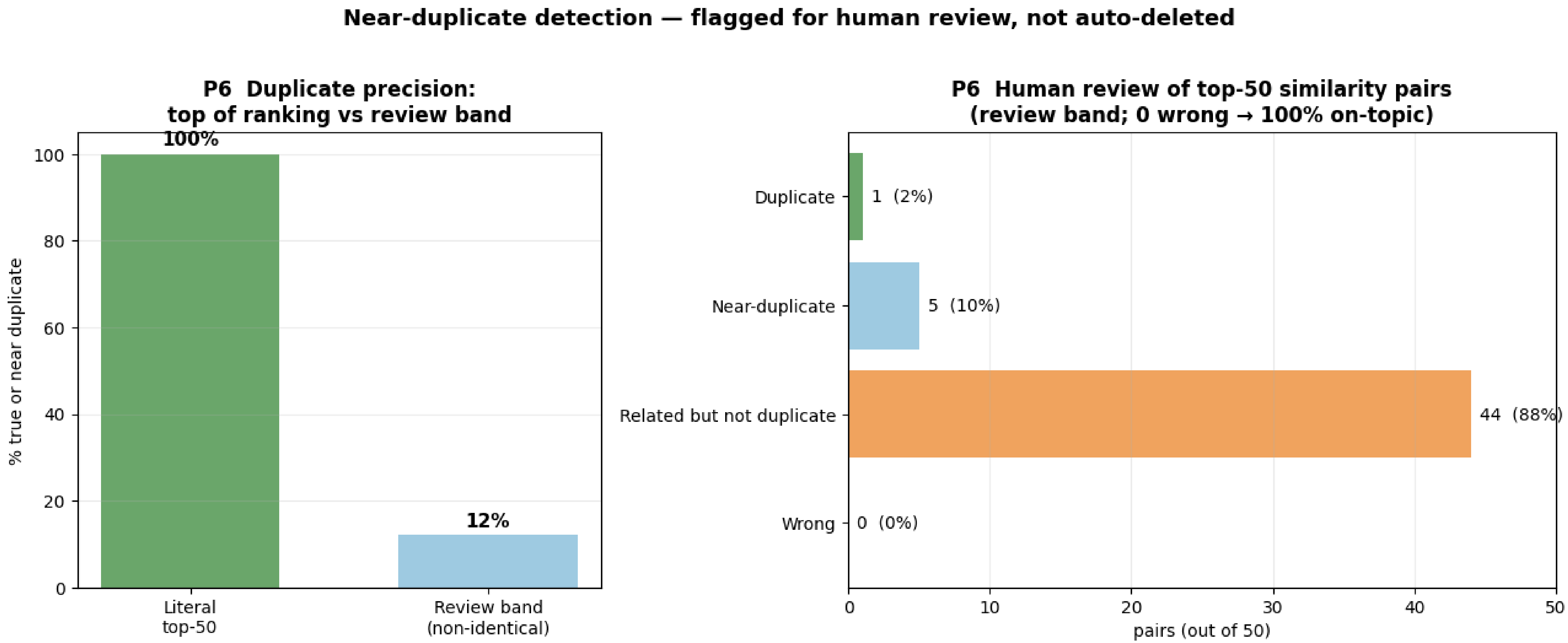


A · Interpretability 71 of 76 themes read as a clear topic

B · Sample consistency 5 random datasets per theme belong together

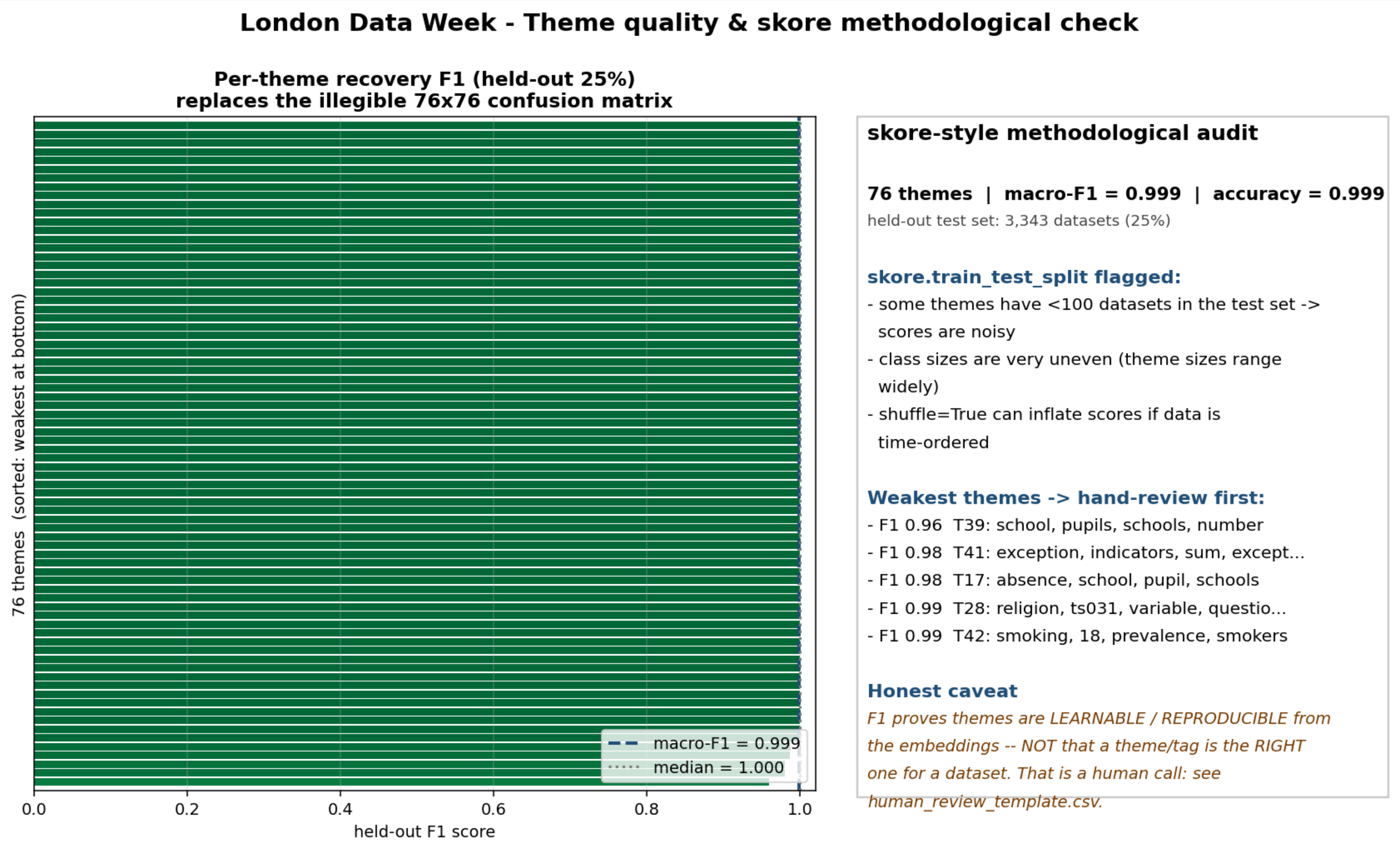
C · Stability themes hold when parameters change (mean ARI 0.841)

Flagged for Review — Not Auto-Deleted



Honest finding: among non-identical high-similarity pairs, only 12% turn out to be true or near duplicates on human review — 88% are just related, not duplicate.

How We Chose — and Checked — Our Method



What We Trust — and What We Don't

We do not claim the AI-generated metadata is automatically correct.

Can trust

- Search: semantic beats keyword on every IR metric (Precision@5 0.97 vs 0.42) — including a synonym it was never told, NEET
- Tags: 93.3% specific and usable as suggestions
- Themes: stable under parameter change (ARI 0.841); genuinely interpretable samples

Not yet

- Duplicates: only 12% of non-identical high-similarity pairs are true/near duplicates — needs human review, not auto-merge
- Themes: still some source (borough) bias in a subset of clusters
- 35% of datasets are honestly left 'unclustered' — real metadata gaps, not hidden

Thank you.

Trustworthy as a decision-support layer — not an unchecked replacement for human metadata curation.